Suggested solutions to examination in Group and Ring Theory 2021-05-28

(1.) We first calculate the Smith normal form of the matrix, over $\mathbb{Z}$,

$$\begin{bmatrix} 7 & -4 & 2 \\ -16 & 10 & -5 \\ 23 & -14 & 7 \\ -14 & 8 & -4 \end{bmatrix} \sim \begin{bmatrix} 7 & 0 & 2 \\ -16 & 0 & -5 \\ 23 & 0 & 7 \\ -14 & 0 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ -1 & 0 & -5 \\ 2 & 0 & 7 \\ -2 & 0 & -4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 0 \\ -1 & -5 & 0 \\ 2 & 7 & 0 \\ -2 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ -1 & -3 & 0 \\ 2 & 3 & 0 \\ -2 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times(-1) \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus there is a set of generators y_1, y_2, y_3, y_4 ~~such~~ of A such that the relations are given by

$$y_1 = 0, \quad 3y_2 = 0$$

Hence

$$A \cong \frac{\mathbb{Z}y_1 \oplus \mathbb{Z}y_2 \oplus \mathbb{Z}y_3 \oplus \mathbb{Z}y_4}{\mathbb{Z}y_1 \oplus 3\mathbb{Z}y_2} \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}.$$

(2.) (i) $\Rightarrow$ (ii) By (i) there are $e \in I$, $f \in J$ such that

$$1 = e + f.$$

Multiply by e on the left.

$$e = e^2 + ef$$

Here $e, e^2 \in I$, $ef \in J$. As $I \oplus J$ is a direct sum it follows that $e = e^2$, $ef = 0$. Let $x \in I$. Then

$$x = x \cdot 1 = xe + xf$$

and as $x, xe \in I$, $xf \in J$, it follows that $x = xe$. In particular $x \in Re$ so that $I \subset Re \subset I$ and hence $I = Re$.

(ii) $\Rightarrow$ (i) Set $f = 1 - e$, $J = Rf$. For $x \in R$,

$$x = x \cdot 1 = x \cdot (e + f) = xe + xf \in I + J$$

so that $I + J = R$. Let $y \in I \cap J$. As $y \in I = Re$ there exists $z \in R$, $y = ze$. As $y \in J = Rf$ there exists $w \in R$ such that $y = wf$. Then $y = ze = zee = wfe = w(1-e)e = w(e - e^2) = w \cdot 0 = 0$. Hence $I \cap J = (0)$ and $I + J$ forms a direct sum.

(3.) We want to define a map

$$f : A \times \mathbb{Z}/m\mathbb{Z} \to A$$
$$(a, [k]_m) \mapsto ka$$

As A has order m, $ma = 0$ for all $a \in A$ by Lagrange theorem. Hence f is well-defined. We check that f is balanced:

$$f(a_1 + a_2, [k]_m) = k(a_1 + a_2) = ka_1 + ka_2 = f(a_1, [k]_m) + f(a_2, [k]_m)$$
$$f(a, [k]_m + [l]_m) = f(a, [k+l]_m) = (k+l)a = ka + la = f(a, [k]_m) + f(a, [l]_m)$$
$$f(na, [k]_m) = kna = nka = f(a, [nk]_m)$$

for $a, a_1, a_2 \in A$, $k, l \in \mathbb{Z}$, $n \in \mathbb{Z}$. By the UMP of the tensor product f induces a group homomorphism

$$\bar{f} : A \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \to A$$
$$a \otimes [k]_m \mapsto ka$$

The map

$$g : A \longrightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z}$$
$$a \longmapsto a \otimes [1]_m$$

is clearly a group homomorphism. We find

$$\bar{f}(g(a)) = \bar{f}(a \otimes [1]_m) = 1 \cdot a = a,$$

$$g(\bar{f}(a \otimes [k]_m)) = g(ka) = ka \otimes [1]_m = a \otimes [k]_m$$

for $a \in A$, $k \in \mathbb{Z}$. Hence $g = \bar{f}^{-1}$ and $\bar{f}$ is bijective and an isomorphism.

4. We show the following slightly stronger statement by induction on n.

<u>Prop.</u> Let $\varphi : \bigoplus_{i=1}^{n} M_i \longrightarrow \bigoplus_{j=1}^{m} M_j'$ be an isomorphism,

where the M_i and M_j' are simple left R-modules. Then $n = m$ and there is a permutation π such that $M_i \simeq M'_{\pi(i)}$, for $1 \leq i \leq n$.

Pf. If $n = 1$, then $\bigoplus_j M_j'$ is isomorphic to M_1, hence simple, so that $m = 1$. Assume true for isomorphisms from direct sums from $< n$ simple modules. As $\bigoplus_{j=1}^{M} M_j' =: C$ is completely reducible, there is by Theorem 14.4.3 a subset $S \subset \{1, 2, \ldots, m\}$ such that $C = \varphi(M_n) \oplus \bigoplus_{j \in S} M_j'$. Let $T = \{1, 2, \ldots, m\} \setminus S$. Then

$$M_n \simeq \varphi(M_n) \simeq \frac{C}{\bigoplus_{j \in S} M_j'} \simeq \bigoplus_{j \in T} M_j'$$

and the simplicity of M_n again forces T to have only one element, $T = \{t\}$.

Then $S = \{1,2,\dots,m\} \setminus \{t\}$. The isomorphism φ induces an isomorphism

$$\bigoplus_{j=1}^{n-1} M_j \simeq \frac{\bigoplus_{j=1}^{n} M_j}{M_n} \simeq \frac{\bigoplus_{j=1}^{m} M_j'}{\varphi(M_n)} \simeq \bigoplus_{j\in S} M_j' .$$

By the induction hypothesis $n-1 = |S| = m-1$, so that $n=m$, and there exists ~~an~~ a bijection

$$\pi : \{1,2,\dots,n-1\} \to S$$

such that $M_i \simeq M'_{\pi(i)}$. Extend π to $\{1,2,\dots,n\}$ by defining $\pi(n) = t$, then $M_n \simeq M_t'$ completes the induction. $\square$

Applying the proposition to the identity isomorphism $id: M \to M$ gives the result.

(5.) Define

$$f : R \to R/I_1 \oplus \cdots \oplus R/I_n$$
$$r \mapsto (r+I_1, \dots, r+I_n).$$

Then f is a homomorphism of left R-modules. Let $x \in \ker(f)$. Then $x + I_k = 0 + I_k$ so that $x \in I_k$ for each $k=1,\dots,n$. Hence $x \in I_1 \cap \dots \cap I_n = (0)$. Hence $\ker(f) = (0)$ and f is injective. As the R/I_k are Noetherian, $R/I_1 \oplus \dots \oplus R/I_n$ is Noetherian by Theorem 19.2.6. As $f(R)$ is a submodule of $\oplus R/I_k$, $f(R)$ is Noetherian by Theorem 19.2.5. As R is isomorphic to $f(R)$, R is also Noetherian.

(6.) Assume for a contradiction that G is a simple group, $|G| = 400 = 2^4 \cdot 5^2$. Let n_5 be the number of Sylow$_5$-subgroups of G. Then $n_5 \equiv 1 \pmod 5$ and $n_5 \mid 16$ so that $n_5 = 1$ or 16 by Sylow's third theorem. If $n_5 = 1$, the unique Sylow$_5$-subgroup is normal in G, contradiction. Hence $n_5 = 16$.

Case 1. There exist two Sylow$_5$-subgroups P, Q with $|P \cap Q| = 5$. Set $H = P \cap Q$. Then $H \triangleleft P$, $H \triangleleft Q$ by Theorem 5.4.10 (iii). Set $N = N_G(H)$. Then $|N| \in \{50, 100, 200, 400\}$. If $|N| \in \{50, 100, 200\}$, then applying Sylow's third theorem to N shows that N has a unique Sylow$_5$-subgroup, contradicting the fact that $P < N$, $Q < N$. Hence $|N| = 400$, $N = G$, $H \triangleleft G$, contradiction.

Case 2. For all pairs P, Q of Sylow$_5$-subgroups of G, $|P \cap Q| = 1$. Then G contains

$$16 \cdot 24 = 384$$

elements of order 5 or 25, the non-identity elements in the Sylow$_5$-subgroups. This leaves only $400 - 384 = 16$ elements. These must make up a unique Sylow$_2$-subgroup S. Then $S \triangleleft G$, contradiction.